Astranesse

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1 Planet generation

1.1 Atmosphere

Given the basic properties of a planet and a rough altitude map the task is to compute its atmospheric flows. These will then be used to determine the weathering and humidity maps of the planet, which modify the altitude and set the biome. The atmospheric simulation should include Hadley cells and respond to large scale features such as coastlines and mountains. The humidity can be done post hoc, and should take adiabatic cooling into account.

1.1.1 Scale height

Force balance requires simply $\nabla P + \rho g = 0$. Note that $g = GM/(w/2)^2 = 4GM/w^2$ where w is the planet width. Writing $P = P_0 e^{-z/H}$, we have $P/H = \rho g$, or $H = P/(\rho g)$. Also note $PV = Nk_BT \implies P = nk_BT \implies P = \rho(k_B/m)T$, where m is the mean molecular mass. Thus, $H = (k_BT)/(mg)$.

1.1.2 Composition

Suppose I know all the molecular species in the atmosphere and their masses M_i. The column density is $M_i/(6w^2)$ where w is the planet width. Since $\rho = \rho_0 e^{-z/H}$, and the column density of this exponent is $\rho_0 H$, that means the surface density satisfies $\rho_i = M_i/(6w^2H)$. It follows that $P_i = H\rho g = M_i g/(6w^2)$. That would be if they were a gas.

But are they a gas? Suppose the solid-liquid phase transition is independent of pressure. So if $T < T_c$, the choice is between gas and solid, and otherwise it's between liquid and solid. In both cases, there's a $P_i^v(T)$ vapor pressure curve. The liquid/solid state is maintained if vapor pressure can equal pressure. So if $P_i^v(T) > P_i$, the molecule is completely a gas. Otherwise, the molecule's partial pressure is its vapor pressure, and the mass in solid/liquid phase is $M_i P_i^v(T)/P_i$.

1.1.3 Temperature

Take a patch of ground has normal vector $\hat{\boldsymbol{n}} = (\sin\theta, 0, \cos\theta)$ in the local frame. This frame rotates around spin pole $\hat{\boldsymbol{z}}$, with sun vector $(0, \sin\phi, \cos\phi)$ in the global frame. This is a general setup; the planet rotates, so the azimuthal angle of $\hat{\boldsymbol{n}}$ can be fixed, and then by global rotational symmetry the spin pole can be set to $\hat{\boldsymbol{z}}$ and the azimuthal angle of $\hat{\boldsymbol{s}}$ to the value I chose. Suppose the solar flux is F and absorption (one minus albedo) is ϵ. Then the injected energy is $F\epsilon dA \max[\hat{\boldsymbol{n}}(t)\cdot\hat{\boldsymbol{s}}, 0]$, where $\hat{\boldsymbol{n}}(t)$ is in the global frame. Over one day, it's $\int_0^{2\pi/\Omega} dt \max[\hat{\boldsymbol{s}}^T R_z(\Omega t)\hat{\boldsymbol{n}}, 0]$. This equation is hard to evaluate. But I could evaluate by analogy: $\int_0^{2\pi/\Omega} dt \max[\sin(\Omega t), 0] = 2/\Omega$, and $\int_0^{2\pi/\Omega} dt \sin(\Omega t)^2 = \pi/\Omega$. Thus, perhaps the integral of this max is approximately $2/\pi$ times the integral of the square? In that case,

$$\Delta E = \frac{2}{\Omega}\epsilon\left[\cos^2\theta(3\cos^2\phi - 1) + \sin^2\phi\right] FdA. \quad (1)$$

Meanwhile, the loss due to radiation is $\Delta E = \sigma T^4 dA(2\pi/\Omega)$. Altogether,

$$T = \frac{3}{2}T_0\left(\frac{\epsilon}{\epsilon_0}\right)^{1/4}\left[1 - (\hat{\boldsymbol{n}}\cdot\hat{\boldsymbol{\Omega}})^2(1 - 3(\hat{\boldsymbol{s}}\cdot\hat{\boldsymbol{\Omega}})^2) - (\hat{\boldsymbol{s}}\cdot\hat{\boldsymbol{\Omega}})^2\right]^{1/4}. \quad (2)$$

where $T_0 = (2/3)[(F\epsilon_0)/(\pi\sigma)]^{1/4}$, and ϵ_0 is the surface-averaged absorption. Note that there's diurnal variation in $\hat{\boldsymbol{s}}\cdot\hat{\boldsymbol{\Omega}}$: it varies sinusoidally over the year. The average value of $(\hat{\boldsymbol{s}}\cdot\hat{\boldsymbol{\Omega}})^2$ over a year is thus the max squared over two. It might also be interesting to know that the average value of T over the full surface is T_0.

That's the temperature of a planet with no atmosphere. But as the atmosphere carries heat, a dense atmosphere will restore the temperature to a uniform T_0. Thus, the true temperature is something more like $\eta T + T_0(1-\eta)$, for $\eta \in [0, 1]$.

What is η? It must relate to how much heat can be transferred by the atmosphere. The scale of heat flux due to the

sun is $\sigma T^4 dA$ for a region of area dA. Meanwhile, the atmosphere has energy density ρCT for specific heat capacity C, and if it moves at speed v then the heat flux is $\rho CTvdA$. Define $\tanh^{-1}\eta$ as the ratio between the convective and radiative heat flux. I will note later that $v \sim \sqrt{k_B T/m} \sim \sqrt{Hg}$. It also might be useful to remember that $\rho = P/(Hg)$. Thus

$$\eta \sim \tanh \frac{PC}{\sigma T^3 \sqrt{Hg}}. \tag{3}$$

1.2 Wind

For the momentum equation, I'll measure the effects of pressure, Coriolis effect, and mountains. No viscosity. For pressure, once this heating is applied to air near the ground the rises when $T > T_0$ and sinks otherwise. This causes a pressure low / high. A sensible guess would be $\Delta P/P_0 = -\Delta T/T_0$, where P_0 is the average pressure. Thus, the force is $f_p = P_0 \nabla T/T_0$. By the ideal gas law, $P_0 = \rho_0 Hg$.

For the Coriolis effect, the force density is $f_c = 2\rho\boldsymbol{\Omega} \times \boldsymbol{v}$ is the Coriolis force per unit mass. And for mountains, a chunk of mass moving up requires work density $\rho \Delta hg$. Thus, the force was $f_m = -\rho hg/\Delta x$, meaning $-\rho g \nabla h$. Kinda weirdly, this prefers atmosphere to roll down the mountain, when that's not really what I wanted. But I'll continue. Momentum balance gives $\rho D\boldsymbol{v}/Dt = f$, but in steady state this is

$$(\boldsymbol{v}\cdot\nabla)\boldsymbol{v} = gH\left(\nabla\frac{T}{T_0} - \nabla\frac{h}{H}\right) + 2\boldsymbol{\Omega}\times\boldsymbol{v} \tag{4}$$

I'll have to update this numerically.

1.3 Fog

Fog consists of the combined effects of scattering out of the line of sight and into the line of sight. Starting with the former (extinction),

1.3.1 Extinction

I'll use the optical depth law

$$\tau = \int_0^\infty d\ell\, E(\ell) = \int_0^\infty d\ell\, E_0 e^{-h(\ell)/H} \tag{5}$$

where $E(\ell)$ is the differential extinction and the fog color is blended into the background color by the amount $1 - e^{-}\tau$. The second equation comes from assuming an exponentially decaying atmosphere with surface extinction E_0 (inverse meters), scale height H, and height $h(\ell)$ above the surface.

For a cubic planet, $h(\ell)$ depends on what face the point of integration is hovering over. So I'll split the trajectory into chunks, where each chunk is over a unique face. One way of formalizing this is to define the vector $\boldsymbol{\ell}$ pointing from the camera to the end of the sight ray, and dividing the vector into chunks ranging from $\alpha_i\boldsymbol{\ell}$ to $\alpha_{i+1}\boldsymbol{\ell}$, so that

$$\tau = |\ell| E_0 \sum_i \int_{\alpha_i}^{\alpha_{i+1}} d\alpha\, e^{-h_i(\alpha)/H}. \tag{6}$$

If the chunk is over a face with normal vector $\hat{\boldsymbol{n}}$, then the height is $(\boldsymbol{a}+\alpha\boldsymbol{\ell})\cdot\hat{\boldsymbol{n}} - w$ where w is the cube half-width and $\boldsymbol{a}$ is the ray's starting position. The integral of a single term is

$$\tau_i = E_0 H \frac{\ell}{\boldsymbol{\ell}\cdot\hat{\boldsymbol{n}}_i} e^{-(\boldsymbol{a}\cdot\hat{\boldsymbol{n}}_i - w_i)/H}\left(e^{-\boldsymbol{\ell}\cdot\hat{\boldsymbol{n}}_i\alpha_i/H} - e^{-\hat{\boldsymbol{\ell}}\cdot\hat{\boldsymbol{n}}_i\alpha_{i+1}/H}\right). \tag{7}$$

It seems that one easy way to continue this formula for edges and corners is just to take the normal vector as the average of the abutting faces' normal vectors.

Note that $E_0 H$ is dimensionless. I call that the extinction constant. Then the coordinate system can be remapped in whatever way is desired, e.g. to a frame in which the planet is the unit cube. Then $w_i = 1$, $\hat{\boldsymbol{n}}_i$ are the unit axes, and H is the only important length scale.

1.3.2 Scattering

The amount of light scattered into the line of sight is rather simply computed. If scattering is isotropic and the radiation density going in direction $\hat{\boldsymbol{t}}$ at point $\boldsymbol{x}$ is $\rho(\boldsymbol{x}, \hat{\boldsymbol{t}})$, then the radiation per unit travel length scattered into solid angle $d\Omega$ is $E(\boldsymbol{x})\rho(\boldsymbol{x},\hat{\boldsymbol{t}})/(4\pi)$, where the 4π factor satisfies that the integral of all the scattered light gives extinction. So the flux of scattered light is

$$I = \int_0^\infty d\Omega_{\hat{\boldsymbol{t}}} d\ell \frac{E(\ell)}{4\pi}\rho(\boldsymbol{x},\hat{\boldsymbol{t}}). \tag{8}$$

What is the radiation density though? If optical depth is small, $\rho(\boldsymbol{x},\hat{\boldsymbol{t}}) = 0$ except along the axis to the sun $\hat{\boldsymbol{s}}$. Specifically, $\rho(\boldsymbol{x},\hat{\boldsymbol{t}}) = I_0 e^{-\tau_0}\delta(\hat{\boldsymbol{t}} - \hat{\boldsymbol{s}})$ where I_0 is the illumination of the sun, which sets the color of each block. I computed this integral for the case where the sun extends only over one face:

$$I/I_0 = \frac{1}{4\pi}\frac{\boldsymbol{s}\cdot\hat{\boldsymbol{n}}|\boldsymbol{\ell}|}{\boldsymbol{\ell}\cdot\hat{\boldsymbol{n}}|\boldsymbol{s}|}\left[\exp\left(e^{-((\boldsymbol{a}+\alpha_{i+1}\boldsymbol{\ell})\cdot\boldsymbol{n} - w_i)/H}\frac{|\boldsymbol{s}|}{\boldsymbol{s}\cdot\hat{\boldsymbol{n}}_i}E_0 H\right) - \exp\right. \tag{9}$$

2 Gravity

I need to make gravity point normal to the planet at the face of the planet. This can be done by warping space according to a metric such that the unit sphere gets mapped to a cube, but points at infinity are fixed. Then I just map gravitational forces using this metric. One example map is

$$\boldsymbol{x} \to \boldsymbol{x}' = \boldsymbol{x} f(\boldsymbol{x}), \qquad f(\boldsymbol{x}) = 1 + \frac{H}{x}\left[\frac{P(\boldsymbol{x})}{x} - 1\right] \tag{10}$$

where H is the radius of the sphere that gets mapped to the cube, $P(\boldsymbol{x}) = \left(\sum_i x_i^p\right)^{1/p}$, and p controls the corner softness ($p \to \infty$ is sharp). What is the metric? The Jacobian is $J_{ij} = \partial_i x_j = \delta_{ij} f + x_i \partial_j f$, and according to Claude the metric is $f = J^T J$. Thus,

$$g_{ij} = f^2 \delta_{ij} + f(x_i \partial_j + x_j \partial_i) f + x^2 \partial_i \partial_j f. \tag{11}$$

The Christoffel symbols satisfy

$$\Gamma^i_{kl} = \frac{1}{2} g^{im} \left(g_{mk,l} + g_{ml,k} - g_{kl,m}\right). \tag{12}$$

Plugging these into the equations of motion, one gets that $\ddot{x}'_i = \ddot{x}_j + \Gamma^i_{jk} \dot{x}_j \dot{x}_k$. Claude handled this for me so that it's completely a function of x'. The result is

$$\ddot{\boldsymbol{x}}' = -\frac{GM\,\boldsymbol{x}}{r^3} + \frac{2H\,(\mathbf{u}\cdot\dot{\boldsymbol{x}}')}{r^2 D}\,\dot{\boldsymbol{x}}' + \frac{H}{r^2 D^2}\left[w_{ij}\,\dot{x}'_i\,\dot{x}'_j + \frac{2H}{r^2}\,(\mathbf{u}\cdot\dot{\boldsymbol{x}}')^2\right]\boldsymbol{x}, \tag{13}$$

$$\boldsymbol{x} = \mathbf{x}'\left[1 - \frac{H}{|\mathbf{x}'|}\left(\frac{P(\mathbf{x}')}{|\mathbf{x}'|} - 1\right)\right], \tag{14}$$

$$r = |\boldsymbol{x}| = |\boldsymbol{x}'| - H\left(\frac{P(\boldsymbol{x}')}{|\boldsymbol{x}'|} - 1\right) \tag{15}$$

$$P = \left(|x_1|^p + |x_2|^p + |x_3|^p\right)^{1/p} \tag{16}$$

$$D = r^2 + H(P - r), \tag{17}$$

$$u_i = r^2 P^{1-p}\, x_i |x_i|^{p-2} + (r - 2P)\, x_i, \tag{18}$$

$$w_{ij} = (p-1)\, r^4 P^{1-p}\left(|x_i|^{p-2}\,\delta_{ij} - P^{-p}\, x_i x_j\, |x_i|^{p-2}|x_j|^{p-2}\right). \tag{19}$$

Now I take $D \to D/r^2$.